Full-Color Quantum Dot Light-Emitting Diodes Based on Microcavities

Guanding Mei, Weigao Wang, Dan Wu, Philip Anthony Surman, Kai Wang, Wallace C. H. Choy, Xiaochuan Yang, Wenwei Xu, and Xiao Wei Sun

Abstract—**Full-color display is a primary challenge for the commercialization of quantum dots (QDs). In this study, we utilize the spectral narrowing phenomenon of microcavities to fabricate the red, green and blue quantum dot light-emitting diodes (QLEDs) with a single QD layer. This work theoretically analyses the role of microcavities in adjusting the emitting color of QLEDs. By enhanced microcavity and properly chosen spacer thickness, the spectral selectivity shifts, realizing the full-color-tunability of QLEDs. The tunable experimental spectra of microcavity QLEDs are observed, in excellent agreement with our theoretical design. Benefiting from the spectral narrowing of microcavity and the narrow spectra of QDs, a high color purity with full width at half maximum (FWHM) of 18 to 25 nm is realized, leading to a color gamut ratio of 104.8% compared to National Television System Committee (NTSC) standard. The light extraction is also enhanced by constructive interference and the Purcell effect in the microcavity. Moreover, in the fabrication of red, green, and blue pixels, patterning the transparent cathode has better feasibility and lower damage relative to patterning the light-emitting layer.**

Index Terms—**Cavity resonators, displays, light emitting diodes, quantum dots.**

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I. Introduction

QUANTUM dot light-emitting diodes (QLEDs) have become a promising electroluminescent (EL) device for next-generation displays, with advantages of solution-processability [1], color-tunable emission with narrow bandwidth [2]–[5] and remarkable EL efficiency [6]–[10]. A typical top-emitting QLED device architecture as plotted in Fig. 1(a) is composed of an anode, hole injection layers (HILs), a hole transport layer (HTL), an emitting layer (EML), an electron transport layer (ETL), an electron injection layer (EIL) and a cathode. To realize full-color and high-resolution QLED displays consisting of red, green and blue (RGB) pixels, EMLs patterning technologies such as inkjet printing [11]–[13], transfer printing [14], photolithography [15], [16] and electrophoretic deposition [17] are widely reported. These technologies manipulate the red, green, and blue QDs to be deposited in spatial positions. Currently, these technologies still have their shortcomings. For example, inkjet printing tends to produce a non-uniform film with a rough morphology, leading to disordered charge transport and charge imbalance [14]; transfer printing may introduce current leakage due to cracks [14]. For photolithography, the QD layer needs to endure multiple photolithographic processes [15]. In this context, it is meaningful to develop a simpler and more cost-effective method to realize full-color QLEDs.

In the present study, we investigate the effect of microcavity in fabricating full-color QLEDs with a white emitter.

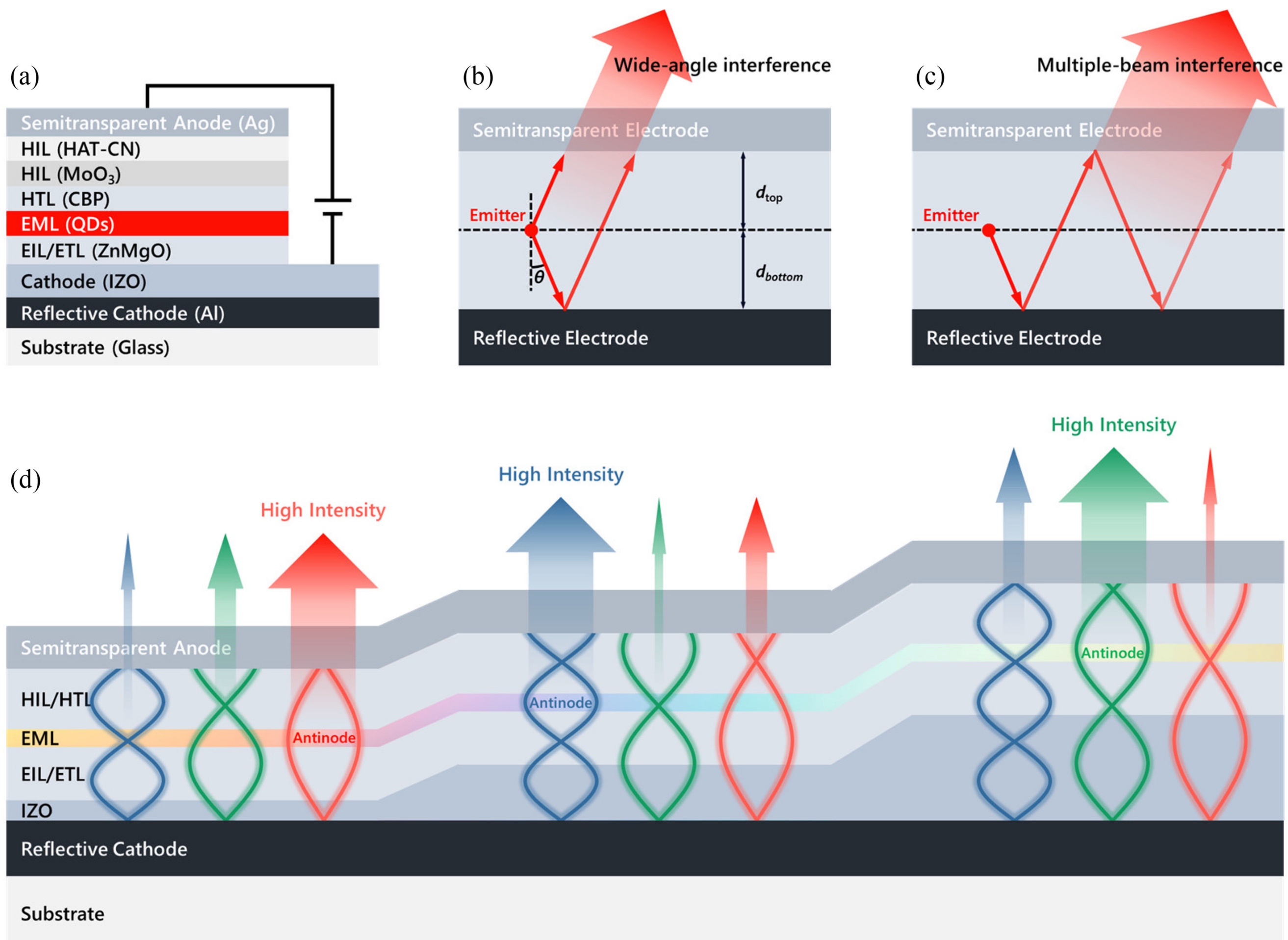

Fig. 1. Schematic device architecture and microcavity effect. (a) A typical device architecture of top-emission QLED. (b) Wide-angle interference and (c) multiple-beam interference in the microcavity. (d) EML is fitted to the antinode positions of the resonance wave of red, blue, and green by varied IZO thickness.

The microcavity effect has been employed to enhance light extraction [18], [19] and to produce narrow spectra [20], [21] in light-emitting diodes (LEDs). Based on its spectral narrowing phenomenon, a thin film color filter with high selectivity can be implemented by a microcavity structure as reported in [22]. In white organic light-emitting diodes (OLEDs), color filters, a distributed Bragg reflector (DBR), and patterned ITO cavity spacers have been used to fabricate the full-color OLEDs [23], in which there is still room to further improve the color purity for display applications. Very recently, the QD hybrid OLEDs [24] and the pixelated QLEDs [25] with cavities for high-resolution displays were also realized. For most bottom-emitting structures, because the ITO spacer is located between the EML and the glass, which has a low reflectance, there is a concern in the modulation of the microcavity. In this work, we examine the top-emitting structure, which is more preferred in the industry as they have a higher aspect ratio. We propose to realize full-color QLEDs by controlling the variable layer (spacer) on top of the bottom reflector for effectively tuning the microcavity, and we will further enhance the microcavity effect by adjusting the top partial transparent electrode for further improving color selectivity.

As shown in Fig. 1(a), a top-emitting inverted structure QLED is sketched. The reflective cathode and multilayer structure of QLEDs indicate that the microcavity is naturally formed in such a confined space due to interference. Therefore, we can utilize the spectral narrowing phenomenon of the microcavity to filter out a single color from the white light emitted by a mixture of red, green and blue QDs. To satisfy color-tunability, we decide to adjust the thickness of the transparent spacer (e.g., indium zinc oxide (IZO) in our case here) to modulate the effective cavity length and the emitter's location. Hence the spectral selectivity of the microcavity shifts, realizing the full-color-tunable QLEDs. It is worth mentioning that the spacer needs to be conductive and should not cause adverse effects on the charge injection and transport.

To improve the color purity, we found that the presence of semitransparent top silver (Ag) anode can enhance microcavity due to its high and adjustable reflection. The Ag anode with a thickness of 5-30 nm not only narrows the spectrum but also enhances light extraction efficiency (LEE). However, further increasing the thickness of the Ag anode will further narrow the spectrum but reduce the efficiency due to limited transmittance.

By optimizing the Ag anode thickness and IZO thickness, the microcavity QLEDs can meet the requirements for full-color displays.

Unlike other patterning technologies, our solution based on the microcavity as a color filter is non-destructive to the morphology and quality of the QD layer because the IZO patterning is predefined before the deposition of the QD layer. Then, a single QD layer is deposited via one-step spin-coating, which is a rapid approach to form a uniform and smooth film. Moreover, the patterning of IZO is based on photolithography, indicating the fabrication with ultra-high pixel densities.

II. Results and Discussion

Due to the reflection inside the QLED, the microcavity introduces wide-angle interference (Fig. 1(b)) and multiple-beam interference (Fig. 1(c)). The wide-angle interference describes the interference between directly emitted and reflected waves. Thus, it can be modulated by the distance between the emitter and the reflective surfaces. The multiple-beam interference describes that the wave presents repetitive reflectance and interference with itself. Thus, it can be adjusted by cavity length. Fig. 1(d) presents a simple schematic of resonant waves of red, green, and blue emission in a weak microcavity. The optimal emitter positions for high LEE are located at the antinodes of the resonance wave [26]. By increasing IZO thickness, the position of EML matches the antinode for red, blue and green sequentially. Thus, the devices are expected to emit red, blue, and green light through varied IZO thickness.

The top-emitting QLED structure in this work consists of Glass/Al (100 nm)/IZO/ZnMgO (40 nm)/QDs (about 10 - 15 nm)/CBP (32 nm)/MoO_3 (8 nm)/HATCN (15 nm)/Ag. Considering the multilayer structure and the luminescence properties of QDs, the oscillating dipole model, based on the Chance, Prock, and Sibley (CPS) theory [27], was applied to design the layer thickness of the QLED stack. We first explore the optical simulations to study the role of microcavity as a color filter. Then the strong microcavity is applied to the QLED with mixed RGB QD as EML. For this structure, experiments are also conducted to verify our microcavity design. Finally, the mechanism and energy dissipation of the microcavity RGB QLEDs will be analyzed.

A. The Role of Microcavity

As shown in Fig. 1(b, c), the microcavity resonance condition [18] is given by

$$\frac{2\pi}{\lambda}2nd_{\text{bottom}}\cos\theta - \phi_{\text{bottom}} = m2\pi \tag{1}$$

$$\frac{2\pi}{\lambda}2nd_{\text{top}}\cos\theta - \phi_{\text{top}} = m2\pi \tag{2}$$

$$\frac{2\pi}{\lambda}2n\left(d_{\text{bottom}} + d_{\text{top}}\right)\cos\theta - \left(\phi_{\text{bottom}} + \phi_{\text{top}}\right) = m2\pi \tag{3}$$

where λ is the wavelength of emission in vacuum; n is the averaged refractive index of the functional layers; d_{top} and d_{bottom} are the distances from the emitter to the top and bottom reflective surfaces, respectively; θ is the incident angle; φ_{top} and φ_{bottom} are the phase shifts for the top and bottom reflections, respectively, which can be solved by the transfer matrix method, and m is the resonance order. Eq. (1) and Eq. (2) describe the wide-angle interference. Eq. (3) describes the multiple-beam interference. According to the equation, resonant wavelength shows a strong dependence on d_{top} and d_{bottom}. Thus, we can alter the resonant wavelength by changing the thickness of a variable layer (spacer) in the cavity.

There are four reasons to choose IZO as the spacer. The first reason is that the upward wide-angle interference (Eq. (1)) is dominant relative to downward wide-angle interference (Eq. (2)) due to the high reflectance of the bottom reflective electrode. Hence the spacer is better placed between the EML and the bottom reflective electrode to adjust d_{bottom}. As shown in Fig. 1(a), IZO can satisfy the demand and modulate upward wide-angle interference and multiple-beam interference simultaneously. Secondly, IZO is a conductive electrode whose thickness variation barely alters the electrical performance through the device stack due to its relatively low resistivity [28]. Thirdly, IZO thickness can be accurately controlled by mature film deposition technologies such as evaporation and sputtering. Last but not least, IZO patterning could be performed before the deposition of the QD layer to avoid damage to the QDs. These advantages of IZO are attributed to the top-emitting structure. Conversely, in the bottom-emitting structure, if IZO is located between EML and substrate, IZO has only limited ability to modulate wide-angle interference due to the low reflection from the substrate. In another case, if IZO is located between EML and the top reflective electrode, patterning IZO may damage the functional layer below.

It should be noted that d_{top} also needs to be optimized to satisfy the resonance condition. If the resonance condition of wide-angle interference and multiple-beam interference mismatch in wavelength, this might result in low efficiency or even multiple peaks in the spectrum. To avoid this problem, d_{bottom} and d_{top} need to simultaneously satisfy Eq. (1-3) for the same resonant wavelength, which means the optimal d_{top} is varied for different IZO thicknesses. However, as discussed, IZO thickness is dominant in altering the resonance relative to d_{top}. Thus, we can still find a d_{top} to balance the resonance condition for different wavelengths. Using a fixed d_{top} for RGB colored pixels is very important as this will avoid patterning of d_{top} layer. In this work, the thickness of CBP is optimized to 32 nm for the appropriate d_{top} for RGB colors.

We simulate the emission spectra at the emission angle of 0 degrees. To investigate the color filtering characteristics of a microcavity on a broad spectrum, an isotropic white emitter, which has a unit intensity in the wavelength range of 400-700 nm, is placed in the middle of EML as an emitter. Initially, the top Ag anode is 5 nm thick with high transmittance, the upward wide-angle interference by the bottom reflection is dominant. Fig. 2(a) shows the emission peak shifts through the entire wavelength range by sweeping the thickness of IZO, showing a wide color tunability. Accordingly, the emission peaks of 618 nm (red), 525 nm (green), and 465 nm (blue) can be realized at IZO thicknesses of 20 nm, 120 nm, and 85 nm, respectively, in good agreement with the schematic shown in Fig. 1(d).

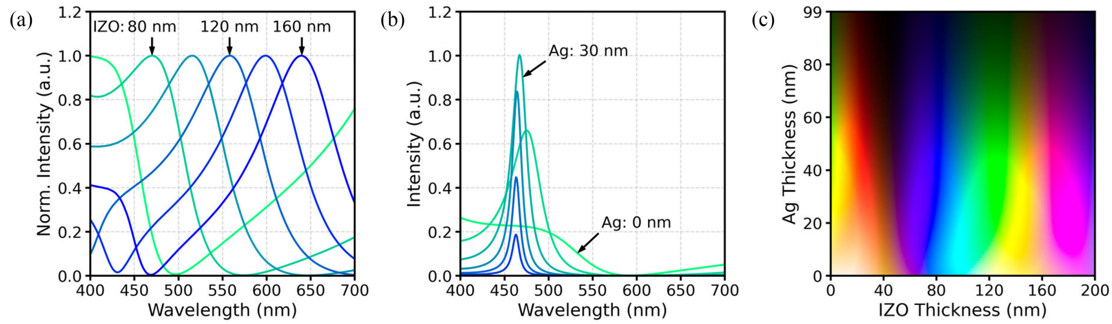

Fig. 2. (a) Spectrum shift of QLED with the white emitter versus IZO thickness variation. (b) Narrowing of the spectral width with thicker top Ag anode at IZO thickness of 85 nm. (c) Emitted colors of QLED with the white emitter versus IZO and Ag thickness variation.

However, the color selectivity is still low according to the wide FWHM of the spectra in Fig. 2(a), which cannot satisfy the color purity requirement for display applications. In order to narrow the spectra and improve the color selectivity, the microcavity needs to be enhanced and precisely controlled. Since the multiple reflections construct multiple-beam interference, a thicker top Ag anode could enhance multiple-beam interference due to stronger reflection. In this case, the structure can be treated as a Fabry−Perot optical resonator, we can define a factor F to evaluate the finesse of the resonator [29], [30], which can be calculated from

$$F \approx \frac{\pi(R_1R_2)^{\frac{1}{4}}}{1-(R_1R_2)^{\frac{1}{2}}} \tag{4}$$

where R_1 and R_2 are the reflectance of bottom and top metal electrodes respectively, given by the Fresnel equations. Larger finesse leads to sharper mode peaks and a larger difference between the minimum and maximum intensity in the microcavity [29], indicating higher color selectivity. From Eq. (4), finesse increases as R_2 increases, which can be enhanced by a thicker top Ag anode. Therefore, we can increase top Ag thickness to improve the color selectivity. One of the reasons we choose Ag as the top anode is that Ag has higher reflectance and lower absorption than aluminum (Al) and gold (Au) in the visible range [31], [32]. This means higher finesse and thus higher color selectivity.

As shown in Fig. 2(b), the spectra are significantly narrowed through a thicker top Ag anode. As the Ag thickness increases from 10 nm to 30 nm, the FWHM narrows from 70 nm to 22 nm. There is almost no red or green light in the spectrum when IZO thickness is 85 nm, and only blue light is emitted from the device. Similarly, the high color selectivity for red and green light can be achieved with properly chosen IZO thickness. This suggests that the microcavity is widely tunable with high color selectivity through the entire visible spectrum. However, if the Ag anode is too thick, it will reduce the transmittance and the light will be dissipated in the multilayer structure, which will be discussed later.

As the simulation of emitted color in the normal direction shown in Fig. 2(c), the variation of IZO thickness and top Ag anode thickness produces an image similar to the hue, saturation and lightness (HSL) color model. This similarity reflects the excellent ability of the microcavity to adjust the emission color. It can be considered that IZO thickness controls the hue of emission, the top Ag thickness modifies saturation, and the drive current adjusts the lightness (brightness). As a result, tuning of IZO and the top Ag thickness in microcavity devices can create most colors across the visible range.

B. Microcavity With Mixed RGB QD

In collaboration with the spectral narrowing of the enhanced microcavity, the intrinsic narrow spectra of QDs also help to achieve superior color purity. Herein, we prepared a mixture of red, green and blue QDs as EML to fabricate the microcavity RGB QLED. The ratio of RGB QDs in the mixture is fixed at 1 : 4 : 17 for obtaining a balanced RGB emission (white emission). The photoluminescence (PL) spectrum of the RGB QD solution is shown in Fig. 3(a). The simulation shows that with the mixed RGB QD as EML, the spectrum at the emission angle of 0 degrees of the microcavity QLED (Fig. 3(a)) can be adjusted to produce RGB emission respectively at IZO thickness of 20 nm, 120 nm and 85 nm. In addition, the FWHM of RGB emission is 25 nm, 18 nm and 20 nm, respectively. These are narrower than the intrinsic PL spectrum of QDs due to the spectral narrowing of the microcavity, indicating a wider color gamut for display.

Compared to Fig. 2(c), the simulation of the emitted color of the mixed RGB QLED in Fig. 3(b) shows sharper color transitions because of the narrow-band emission of QDs. For example, there is no pure yellow emission in the spectrum of the RGB QD mixture (Fig. 3(a)); the yellow emission in Fig. 3(b) comes from the mixture of green and red emissions, as the Ag anode thickness increases, the yellow disappears. This phenomenon implies that, in a strong microcavity (thicker Ag anode), a higher requirement for precise control of IZO thickness is needed to obtain the target color. Usually, evaporation or magnetron sputtering can satisfy the fine-tuning requirement of the IZO thickness.

Derived from the spectra, we plot the CIE color diagram as shown in Fig. 3(c). In a weak microcavity with the top Ag thickness of 5 nm, mixed RGB QLED shows a small color gamut (blue solid lines) with the coverage of 48.4% Adobe RGB [33]

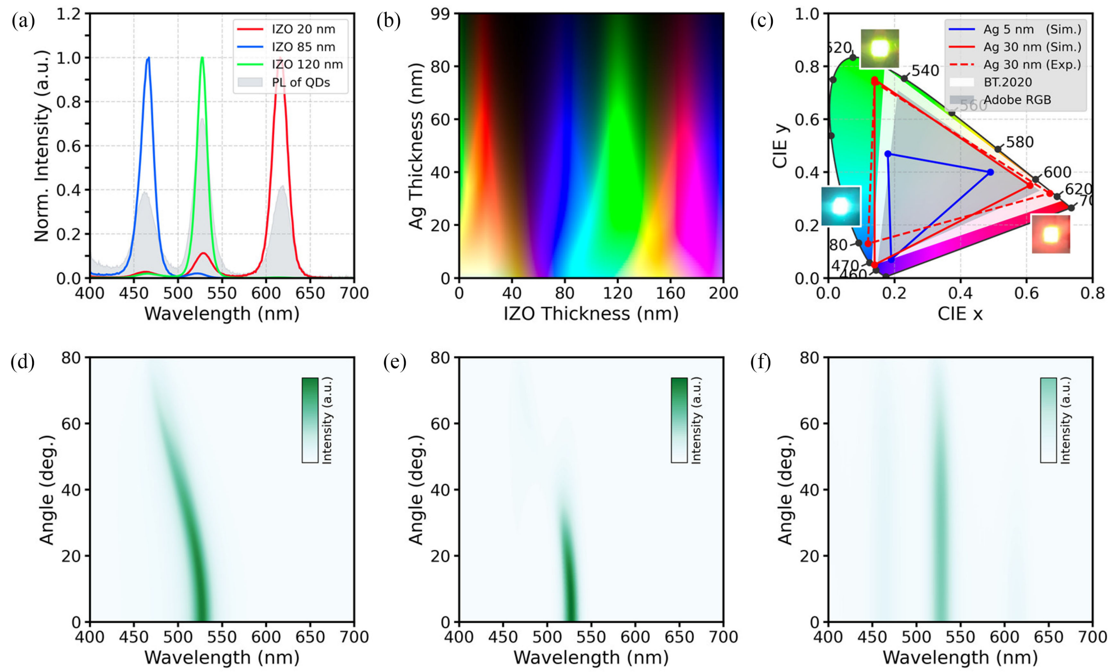

Fig. 3. (a) Photoluminescence spectrum of the RGB QD solution, and the spectra of QLEDs with RGB QD emitter for IZO thicknesses of 20, 85 and 120 nm. (b) Emitted colors of QLED with RGB QD emitter versus IZO and Ag thickness. (c) CIE coordinates of RGB QLEDs with top Ag anode thicknesses of 5 (simulated) and 30 nm (both simulated and experimental). The angular color shift of QLED with 120 nm IZO, for (d) the white emitter and 30 nm Ag, (e) the RGB QD emitter and 30 nm Ag, and (f) the RGB QD emitter and 5 nm Ag anode.

(gray area). While in a strong microcavity with the top Ag thickness of 30 nm, RGB QLED can reproduce up to 91.1% gamut coverage, and 108.8% gamut ratio (red solid lines) of Adobe RGB is achieved. The strong microcavity also leads to a 74.3% gamut coverage of ITU-R Recommendation BT.2020 (white area), which describes the most rigorous color gamut standard for ultra-high-definition television (UHDTV) [34]. This wide color gamut is attributed to the spectral narrowing of microcavity and narrow-band emission of QDs.

Admittedly, the strong microcavity inevitably aggravates angular color shifts [35]. We take the green emission QLED with the IZO thickness of 120 nm as an example. For the strong microcavity with 30 nm top Ag anode, Fig. 3(d) shows the angular intensity distribution of QLED with the white emitter. The spectrum shifts toward a shorter wavelength as the viewing angle increases from 0° to 80°. When the RGB QD mixture is applied (Fig. 3(e)), we see limited color shifts but obvious brightness decay; because of the narrow emission spectra of QDs, there are almost no adjacent colors to shift. On the contrary, Fig. 3(f) shows the weak microcavity with 5 nm top Ag anode for the mixed RGB emitter device, a Lambertian angular distribution is observed, but the color selectivity is poor. Hence, for the microcavity QLEDs, high color selectivity and wide angular distribution of intensity are contradictory.

To verify the microcavity design, we fabricated QLED devices with the structure of Glass/Al (100 nm)/IZO/ZnMgO (40 nm)/QDs (about 10–15 nm)/CBP (32 nm)/MoO_3 (8 nm)/HATCN (15 nm)/Ag (30 nm). The thickness of IZO is set to 90, 120, and 20 nm, which corresponds to the simulated resonant wavelength of 465, 525, and 618 nm, respectively. The spectra of microcavity RGB QLEDs in the normal direction are shown in Fig. 4; the solid curves show the simulated spectra, the scattered ones are the EL spectra detected from the QLED devices at a driving voltage of 6 V. The experimental results agree well with our theoretical analysis for both emission peak wavelength and FWHM; the spectral difference between the simulation and the experiment is within 5 nm, demonstrating the feasibility of our microcavity structure design.

However, there is a slight difference between the experimental and theoretical color purity. In Fig. 4(a), the experimental spectrum shows a higher intensity than the simulation at the wavelength of 525 nm (green), resulting in a slight decrease in color purity for the blue color. On the contrary, in Fig. 4(c), the purity of red emission in the experiment is higher compared to its simulation. To explain this phenomenon, the energy transfer from wide- to narrow-bandgap QDs is considered. We think that in the strong microcavity with 30 nm Ag, multiple reflections provide a relatively high probability of high-energy photons being absorbed by narrow-bandgap QDs. In this case, part of the blue light can be converted into green and red light, and similarly, the green light will also become red light. This re-absorption process belongs to long-range radiative energy transfer, which is not included in the CPS algorithm of the simulation. Moreover, because RGB QDs are closely packed as a thin film, Förster resonant energy transfer (FRET) may also provide a channel [36]–[39]. The FRET between QDs barely exists in the PL spectrum of QD solution (shaded region in Fig. 4), as the input of simulation. Thus, FRET is probably the primary cause for the difference. A recent work [38] demonstrated that the FRET among red-QDs can be suppressed by blue-QDs as spacers. This

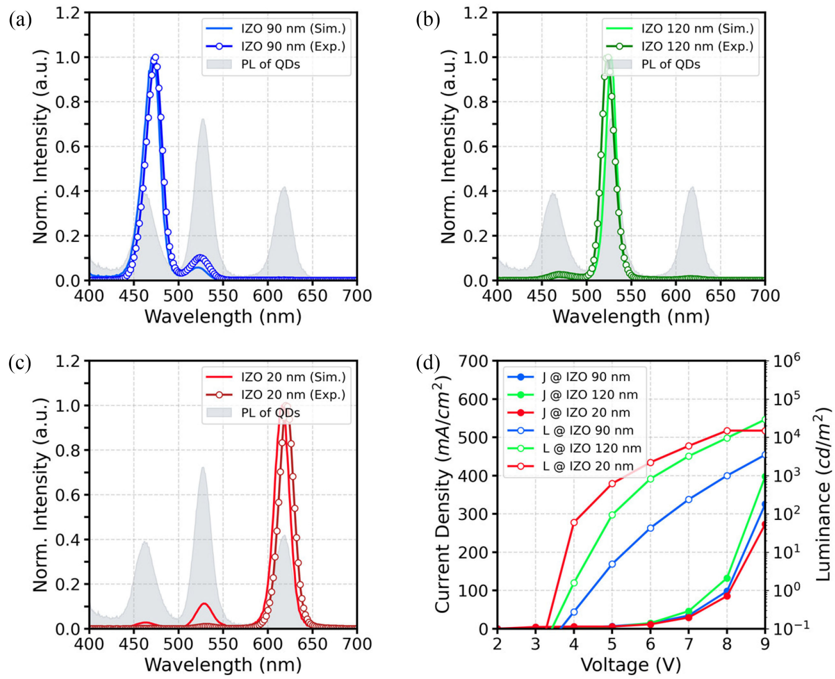

Fig. 4. Comparison between simulated spectra (curves) and experimental EL spectra (scatters) of the RGB QLEDs with IZO thickness of (a) 90 nm, (b) 120 nm, and (c) 20 nm. (d) The current density-voltage-luminance (J-V-L) of the RGB QLEDs.

implies that FTIR will be suppressed to different degrees for different color quantum dots depending on the ratio of RGB QDs.

In Fig. 3(c), the photos of RGB QLEDs are added, and the CIE color coordinates and color triangle of the experimental emission are also added as the red dashed line. Compared with the simulated CIE color diagram, the red primary color is extended due to the energy transfer, but the blue primary color has a slight redshift. Accordingly, 78.3% gamut ratio and 72.3% gamut coverage of BT.2020, as well as 104.8% gamut ratio and 86.9% gamut coverage of national television system committee (NTSC) are achieved. We believe that by further optimizing the microcavity and the composition of the QD mixture, the redshift issue can be addressed, and a wider color gamut can be achieved.

In addition, the current density-voltage-luminance curves are shown in Fig. 4(d). In the normal direction, the luminance ranging from 0–14970, 0–29250 and 0–3512 cd m^{-2} for the R, G and B device, respectively, which can suffice for most display applications.

C. Optical Modes and Energy Dissipation

The enhanced microcavity is also related to LEE. Based on CPS theory, LEE at $\lambda = 525$ nm is shown in Fig. 5(a) with varied IZO and top Ag anode thickness. As stated, the IZO thickness optimized for green color purity is 120 nm, in which the LEE is 18.76%. However, the IZO thickness optimized for LEE of green light is 130 nm, in which the LEE is 22.57%. Similarly, this value for $\lambda = 618$ nm is 27 nm and for $\lambda = 465$ nm is 96 nm. This mismatch between color purity and LEE is due to the narrow angular distribution from the microcavity. As shown in Fig. 5(b), the emission for the 120 nm IZO case is limited at a small angle, while for 130 nm IZO, the emission peak appears at 35 degrees. Although 120 nm IZO shows stronger emission at 0 degrees, the integrated intensity over half space may not be larger than the 130 nm IZO case. In practice, depending on the applications (narrow or wide emission profiles), the thickness of the cavity enhancer (Ag anode) and the cavity spacer (IZO) needs to be carefully chosen.

Another interesting phenomenon is that, despite the weaker transmittance, the increase of the Ag anode thickness does not immediately reduce LEE at peak intensity wavelength. Fig. 5(c) shows the optical power distribution analysis for 525 nm wavelength versus the Ag thickness at 130 nm IZO. The air modes (i.e., LEE) increase as the Ag thickness increase from 0 to 8 nm, then start to decrease as the Ag thickness further increases. The air mode reaches its peak of 34.22% when the Ag thickness is 8 nm, representing a 1.7-fold enhancement compared to 20.08% at the Ag thickness of ideally 0 nm. This is attributed to constructive interference at small angles within the escape cone. Compared with the dual enhancements of color selectivity and efficiency when the Ag thickness increases from 0 nm to 8 nm, the color selectivity improvement with the Ag thicker than 8 nm has the cost of a reduced LEE. In Fig. 5(c), we can also find that the reduced energy is dissipated in absorption and surface plasmon polariton (SPP) modes. Even so, the LEE at 30 nm Ag is 22.57%, which is still higher than 20.08% for the weak microcavity without the top Ag anode. This phenomenon illustrates that color purity and efficiency can be improved simultaneously by the microcavity.

To further illustrate the mechanism, Fig. 5(d, e) show power dissipation spectra based on the white emitter and 130 nm IZO,

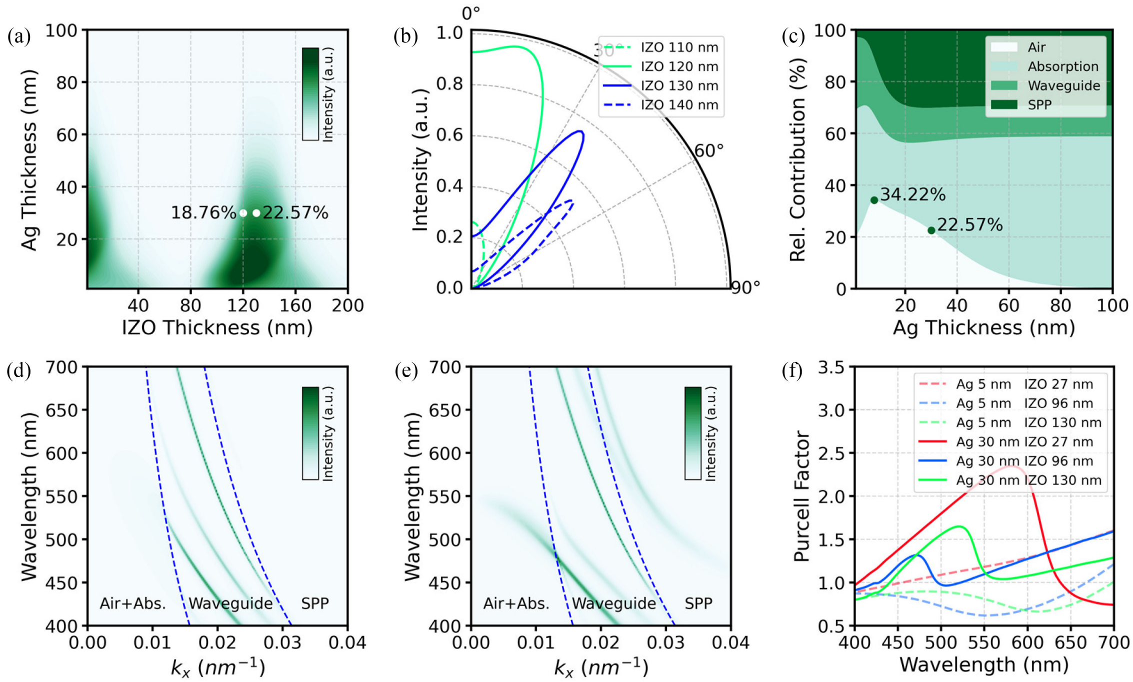

Fig. 5. (a) Air modes at the wavelength of 525 nm versus IZO and Ag thickness variation. (b) Intensity angular distributions at the wavelength of 525 nm with various thicknesses of IZO. (c) Optical power distribution of different optical modes versus Ag thickness at IZO thickness of 130 nm and the wavelength of 525 nm. Power dissipation spectra with the white emitter and 130 nm IZO of (d) weak microcavity QLED with 5 nm Ag anode and (e) strong microcavity QLED with 30 nm Ag anode. (f) Purcell factors versus IZO and Ag thickness variation.

simulated at 5 nm Ag, 30 nm Ag, respectively. By carefully examining these two figures, we can observe that Fig. 5(e) shows more green light in the escape cone ($k_x \in [0, k_0 \cdot n_{\mathrm{air}}]$, where k_x is the in-plane wavevector, $k_0 = 2\pi/\lambda$ is the vacuum wave number, n_{air} is the refractive index of air, $k_x = k_0 \cdot n_{\mathrm{air}}$ is plotted as the left dashed curve), while most of blue and red light is trapped in waveguide modes ($k_x \in [k_0 \cdot n_{air}, k_0 \cdot n_{\mathrm{EML}}]$, where n_{EML} is the refractive index of EML, $k_x = k_0 \cdot n_{\mathrm{EML}}$ is plotted as the right dashed curve) and evanescent modes ($k_x \in [k_0 \cdot n_{\mathrm{EML}}, \infty]$). In Fig. 5(e), we can clearly notice that the waveguide mode smoothly transforms into air mode, implying that the strong microcavity brings waveguide characteristics into air modes, consistent with the angular color shift in Fig. 3(d). These results indicate that the microcavity's excellent color filtering characteristics are not only from the selective trap of light but also from the constructive interference in the target spectral range.

According to Fig. 5(c), when the Ag thickness increases from 5 to 30 nm, more energy is dissipated in the absorption and SPP modes. First, the thicker Ag as an absorber brings more absorption, and as a reflector brings multiple reflections, resulting in multiple absorptions ($k_x \in [0, k_0 \cdot n_{\mathrm{air}}]$). Second, compared with Fig. 5(d), Fig. 5(e) presents stronger evanescent intensity ($k_x \in [k_0 \cdot n_{\mathrm{EML}}, \infty]$), which is attributed to the limited distance between EML and the Ag anode. In this case, the dipole field that overlaps with the evanescent SPP field could be coupled to SPP modes in the optical near field. Before the Ag thickness reaches 30 nm, the constructive interference can still resist the increase in loss. As the Ag anode thickness further increases, the absorption and SPP modes dissipate much optical energy and limit the efficiency. Therefore, it is also essential to balance the color selectivity and the LEE in the microcavities.

A vital efficiency limitation of the mixed RGB QLED is that only 1/3 photons are extracted as 2/3 photons are trapped by microcavity ideally. Fortunately, re-absorption and the Purcell effect [40] can relieve this limitation. First, the blue light trapped in the waveguide modes can be absorbed by green and red QDs, and trapped green light can also be absorbed by red QDs. Thus, trapped blue light can be converted to green and red light, and trapped green light can be converted to red light. This re-absorption process recycles part of the energy trapped by the microcavity color filter. Second, strong microcavity and SPP also enhance the Purcell effect, which can selectively stimulate or inhibit the spontaneous emission rate of QDs. We simulated the Purcell factors for thin and thick Ag top anode, as shown in Fig. 5(f), where the IZO thickness (130 nm) is optimized for LEE. It can be seen that a 30 nm thick Ag produces wavelength-dependent Purcell factor, while a relatively low wavelength-dependence of Purcell factor is observed at 5 nm Ag. An example of the Purcell effect is that at an IZO thickness of 130 nm, the green QDs exhibit a reduced dipole lifetime and thus have an improved rate of spontaneous emission. In this case, more charge carriers recombine to emit green light instead of red and blue. Third, as discussed, LEE improved by strong microcavity also helps to improve the total efficiency. If the structure design can be further optimized, we believe that combining these effects can reduce the efficiency gap between the microcavity mixed RGB QLED and the common single-color QLED.

III. Conclusion

In summary, this study investigated the spectral narrowing characteristics of microcavities in QLEDs and presented a full-color QLED design based on microcavity. With varied IZO thickness, the QLEDs based on a single EML can emit red, green and blue light, respectively, realizing the full color-tunability of QLEDs. The top Ag anode plays an important role in enhancing the microcavity; a top Ag anode of suitable thickness enables QLEDs to achieve the wide color gamut required for displays. Furthermore, the constructive interference, re-absorption, and Purcell effect allow QLEDs to unlock the efficiency limitation of the microcavity mixed RGB QLED. In our solution, to fabricate RGB pixels, only a single mixed QD layer needs to be deposited, and the damage to QDs can be avoided because IZO is pre-patterned. This strategy provides a simple and low-cost solution to realize RGB QLEDs with a wide color gamut and high pixel density, and it indicates the commercial potential of the microcavity RGB QLEDs as well as other types of microcavity RGB LEDs.

Appendix

A. Optical Simulation

Because QD can be approximated as a point dipole source [41], the optical simulations are based on the dipole model and performed by commercial software Setfos and FDTD Solutions. Because the EML (QDs) is only 10–15 nm, it is rational to set the emitter in the middle of the EML. The complex refractive indices of each layer are used to ensure the accuracy of the simulation. The power dissipation spectrum evaluates the power ratio radiated to each in-plane wavevector k_x, which is defined as:

$$k_x = 2\pi n_i \sin\theta_i/\lambda \tag{5}$$

where n_i is the refractive index of i-th layer, θ_i is the propagation angle in i-th layer, λ is the wavelength in vacuum. The rigorous definition of optical channels (air, substrate, waveguide, evanescent, and absorption modes) can be found in the reference [42].

The Purcell effect illustrates that the spontaneous emission rate of emitters can be modified by their local environment; the Purcell factor $F(\lambda)$ is the emission rate enhancement. Therefore, in the microcavity, the effective radiative quantum efficiency (η) is calculated as: [43]

$$\eta = \frac{F(\lambda)\,\Gamma_{\text{rad}}}{\Gamma_{\text{nrad}} + F(\lambda)\,\Gamma_{\text{rad}}} \tag{6}$$

where Γ_{rad} is the radiative decay rate, and Γ_{nrad} is the nonradiative decay rate.

B. Device Fabrication and Measurement

The glass substrates were cleaned by ultrasonication in deionized water, acetone, and isopropanol for 20 min and then dried at 65 °C in a baking oven. The treated substrates were transferred into a high-vacuum evaporation chamber, and 100 nm Al electrode was deposited. Through controlling the operation time, IZO of a certain thickness was deposited by magnetron sputtering. After a 20-min oxygen plasma treatment, the substrates were transferred into a nitrogen-filled glove box, 40 nm ZnMgO nanoparticles (dissolved in ethanol, 20 mg/ml) were spin-coated at 2500 rpm for 45 s and subsequently annealed at 120 °C for 20 min. 10–15 nm QD film (CdSe/ZnS QDs dissolved in normal octane, 10 mg/ml) was spin-coated at 3000 rpm for 45 s and subsequently annealed at 100 °C for 5 min. Then, CBP (32 nm), MoO_3 (8 nm), HAT-CN (15nm), and Ag (30 nm) layers were deposited using a thermal evaporation system under a high vacuum of $<5\times10^{-4}$ Pa. The devices were driven by the DC power source (Keithley 2400, Tektronix, Inc.), and the EL spectra were measured by the spectrometer (PR670).

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